

Built at the MTSU Great Robotics Lab. Every robot the lab owns reports into one platform; agents on it harvest, filter, label and train; a third agent watches the other two. This is what runs today, what the robots have recorded, and what six months of running it taught us.

4

projects on one platform
one ingest contract, one dataset registry

0.8755

our weed detector, mAP50-95
3 seeds, on a sealed 1,977-image test set

15

rounds the agents ran alone
nobody in the room, on the cluster

0.70 vs 0.27

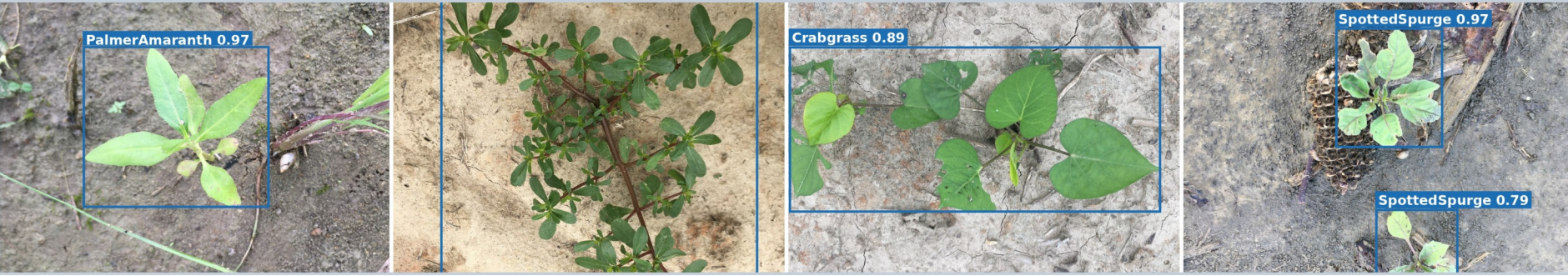
which reviewer cites real evidence
27 B against 7 B, on 149 frozen cases

+0.0287

the cause we found, and tested
and the change the next campaign makes

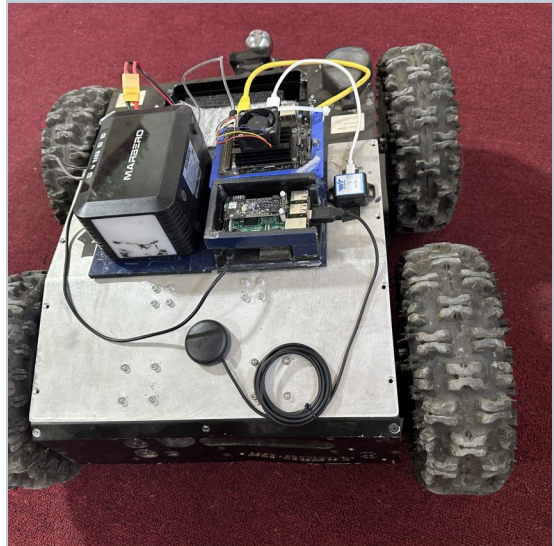
FOUR PROJECTS, ONE CONTRACT; THREE ARE SENDING DATA

Weed Detection



A detector for twelve weed species in cotton. Twenty-seven models were trained or benchmarked against one sealed 1,977-image holdout, and the smallest that held its accuracy is the one the robots carry: 2.6 M parameters, running on live frames. What it cannot do yet is report recall on those frames, because none of them is labelled.

241 Robot



A tracked rover that drives crop rows and streams 640 x 360 camera, GPS and two IMUs straight into the platform: 2,238 frames over 17 drives and 342 m of track. A signed-in user opens its project and drives it from a plain browser, and 1,389 joystick commands have gone through that path. It has been off the network since 2026-08-28, and the platform answers a drive request today with its offline page.

Laser Cart



A galvanometer laser on a wheeled frame, with one camera down over the work zone and one looking forward. It has sent 427 frames over 8 drives, each batch carrying laser state, vehicle telemetry and the cart's own detections. The platform records the cart; its own operator drives it. There is no drive relay, no position sensor, and the platform has never fired the laser.

Humanoid and Quadruped



A humanoid and a quadruped, both out on the farm with the team. The project is open and takes the same ingest contract the rover and the cart use. Nothing either of them recorded has reached the platform yet.

Figure 1. Four projects on one platform. Each was opened the same way, sends through the same ingest contract, and is trained and evaluated by the same agents.

HOW A ROUND RUNS, AND WHAT IT DOES WHEN IT IS STUCK

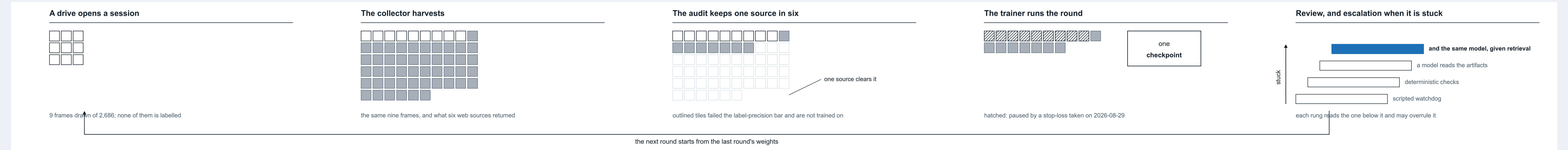
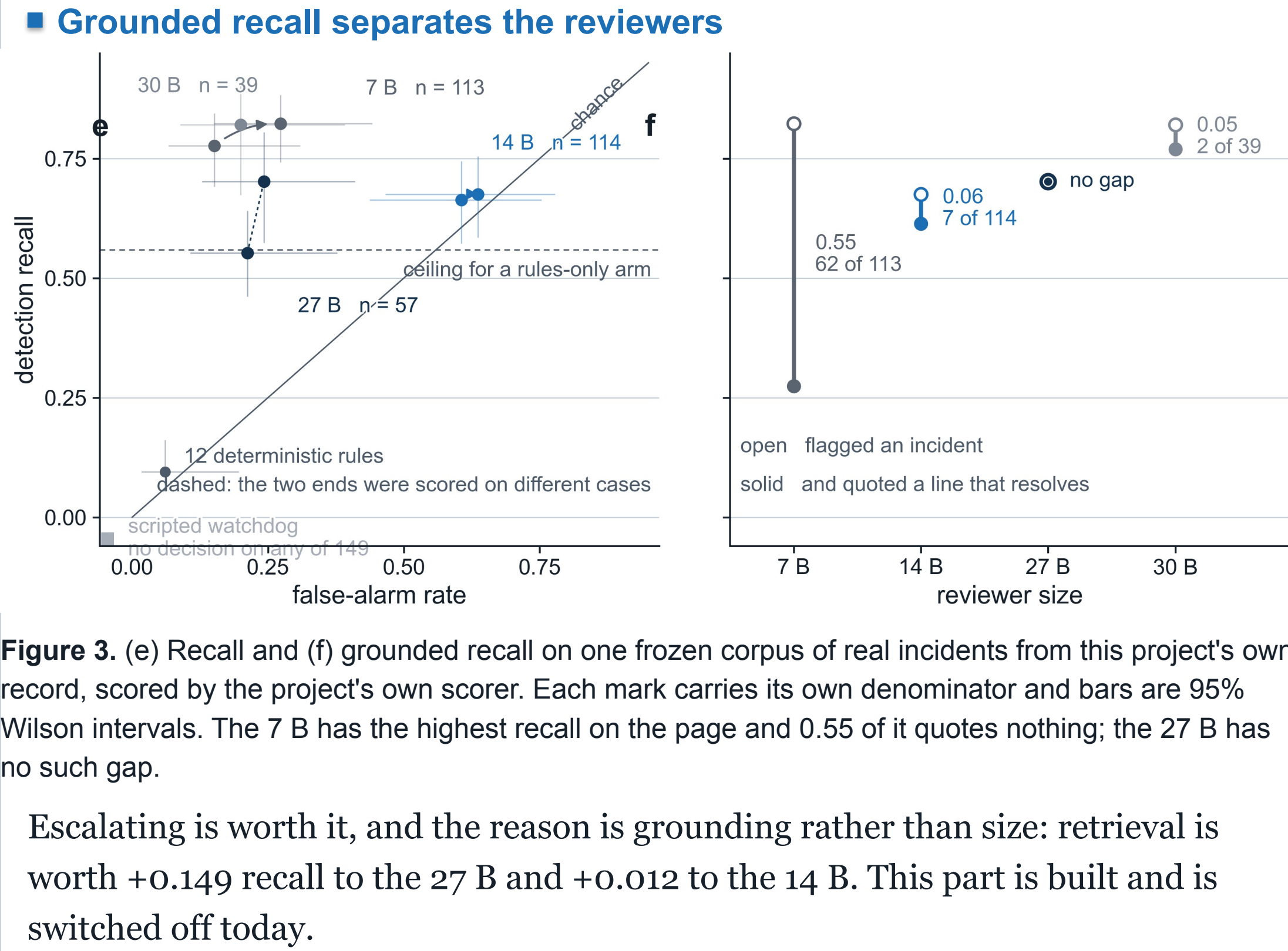


Figure 2. One round, as states. Each column redraws the field of the column before it and changes only what that step changed: nine of the robot's own frames, then what the collector harvested, then the same field with everything that failed the label-precision audit left as an outline, then the round that turns what is left into a checkpoint. Review is a staircase because rank is the point -- each rung reads the one below it. Every stage has run; the collector and the trainer are hatched because a stop-loss paused them on 2026-08-29, and what the loop does not yet do is decide for itself when to climb.

ONE PLATFORM, ANY ROBOT, ANY DATASET

- agentAI is one platform for every robot the lab owns
A robot, a drone or a sensor rig is a project. It owns its keys, its people and a locked dataset registry, and the agents harvest, filter, label and train against that registry. Adding a robot is a configuration change.
- Weed detection is one project here, the humanoid is still empty
Open a project for any data type and any field, upload, and get an automatic diagnosis before choosing what to do next. Weed detection is one project here; so are the rover and the cart.
- Tracked rover and laser cart are the two robots reporting in
Robot 241 is a tracked rover that streams camera, GPS and two IMUs. The laser cart carries a galvanometer laser, a down camera over the work zone and a forward camera.

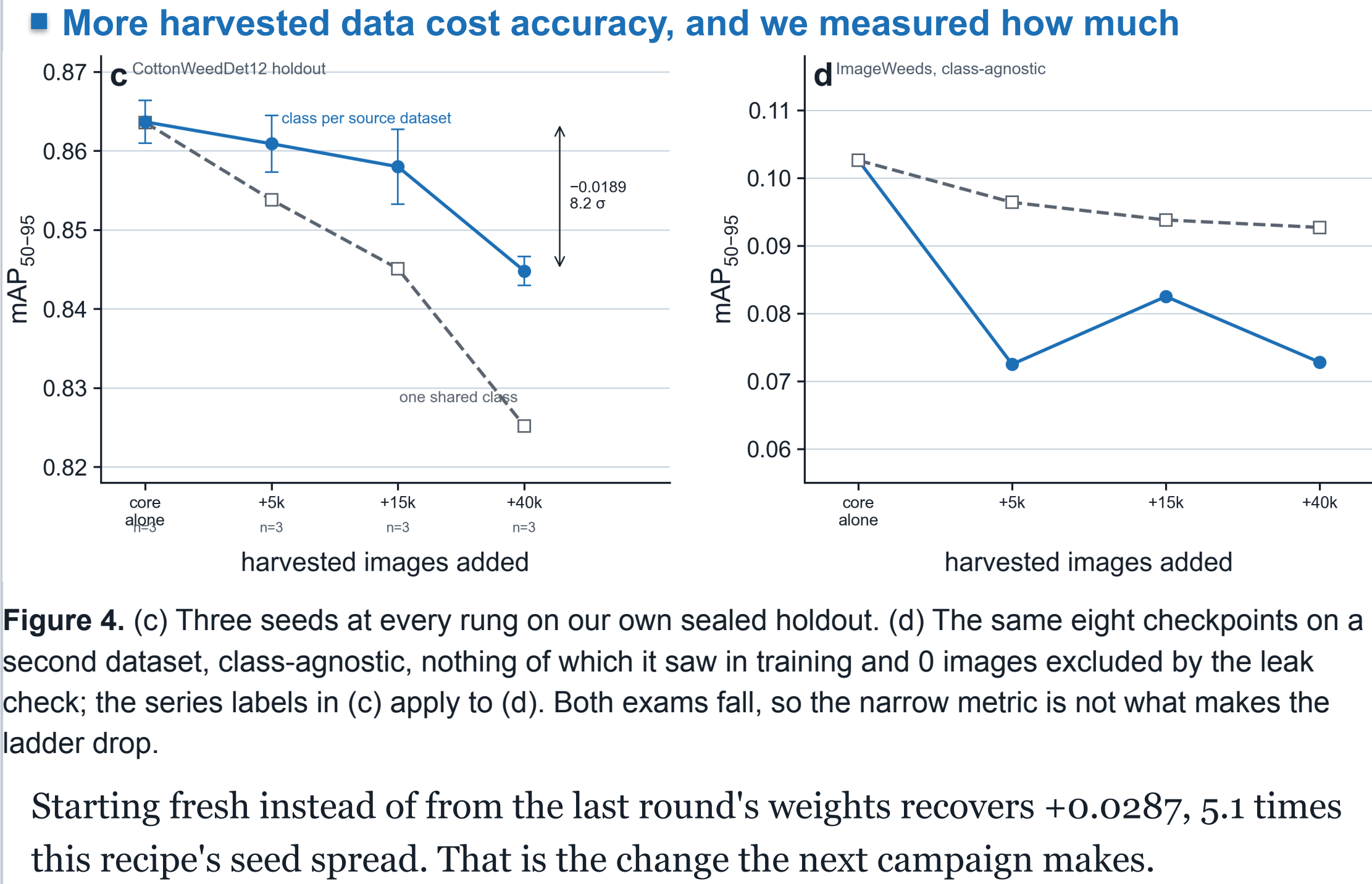
A BRAIN THAT GETS STUCK AND ASKS UPWARD



THE AGENTS RAN THE LOOP WITH NOBODY IN THE ROOM

- Fifteen rounds ran with nobody in the room
Two cluster jobs, one collecting and one training, ran fifteen rounds against the locked registry with nobody in the room. Every round reported success. This part is built and is switched off today.
- The corpus stopped growing at round three
The corpus stopped growing at round three: the same 24 datasets went in every time, and the apparent growth was one staging directory nobody cleared.
- Each finished drive is diagnosed without anyone asking
Every finished drive is diagnosed without being asked -- route, noise, cross-sensor correlation, a timeline -- for 26 live robot sessions and 45 charts so far.

WE MEASURED THE CAUSE, AND IT IS FIXABLE



END STATE IS ONE PLATFORM FOR ANY DATASET AND ANY ROBOT

The end state is one general platform: any dataset, any research field, any robot. Agents on it run without being asked, iterate on their own output, escalate to a larger model when they are out of their depth, and are fenced in a sandbox while they do it. Every piece of this poster is a step in that direction. This is the direction, not a description of what runs today.

CAPABILITIES LAND IN THE PLATFORM, SO THE NEXT STUDENT INHERITS THEM

Each capability is built into the platform itself and stays there after the experiment that needed it. The routing table, the ingest contract, the dataset registry, the sandbox and the scorer are all part of the product, so the next student inherits them working. That is the difference between a lab that has done an experiment and a lab that has built somewhere to do the next one.